

The linear Paulsen bound: a streamlined proof

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Abstract

Every real ε -nearly equal-norm Parseval frame of n vectors in $\mathbb{R}^d$ is within squared Frobenius distance $C\varepsilon d$ of an equal-norm Parseval frame, where C is a universal constant. This is a reorganised and substantially simplified version of a recent proof of this bound (produced with an AI system and formalised in Lean). One deterministic engine—a static row-scaling theorem driven by a half-set hitting time (“barrier constant”) of the squared-Gram random walk—is fed by two random seeds: a filtered Gaussian tangent with a deterministic drift when $d \gtrsim \log n$, and a row-tangent Gaussian when $n \gtrsim d^2$. Compared with the original manuscript: (i) the filter bias is cancelled by a deterministic drift instead of a second scaling, which removes the local correction step, the global spectral comparison, the concentration of squared Gaussian graphs and the matrix-series bound; (ii) a centred remainder estimate lets the many-row seed work at every error level below an absolute constant, which removes the random block partition and the d^6 block size; (iii) the Schur-complement Poisson estimate is replaced by an explicit barrier. All probabilistic estimates are proved in full.

Contents

1	The result and the plan	2
1.1	Statement	2
1.2	The plan	3
2	The scaling toolbox	4
2.1	Projections, distances, and the squared-Gram Laplacian	4
2.2	Diagonal scaling	5
2.3	The barrier constant and static balancing	5
2.4	Barriers from a dense core	6
2.5	From a seed to an exact frame	7
3	Bounded rank: the Hamilton–Moitra estimate	8
4	The many-row seed	9
4.1	The noise	9
4.2	Row normalisation and the exact second-order identity	10
4.3	A dense reference graph	11
4.4	Whitening	11
4.5	Proof of Theorem 4.1	11

5	The moderate-row seed	12
5.1	The horizontal tangent space and the filtered covariance	13
5.2	Commutator identities	14
5.3	The second-order mean and the drift	14
5.4	The expected graph	15
5.5	A good sample	15
5.6	The drifted retraction	16
5.7	Proof of Theorem 5.1	18
6	Assembly	18
A	Gaussian tools	19
B	Relation to the original manuscript and its formalisation	19

1 The result and the plan

1.1 Statement

A frame of n vectors in $\mathbb{R}^d$ is a matrix $X \in \mathbb{R}^{n \times d}$ whose i th row is x_i^T . Distances are *squared* Frobenius distances $\|X - Y\|_F^2 = \sum_i \|x_i - y_i\|^2$, with no normalisation. Throughout,

$$a = d/n.$$

X is *Parseval* if $X^T X = I_d$ and *equal-norm* if $\|x_i\|^2 = a$ for all i ; an equal-norm Parseval frame is an *ENP frame*. X is an ε -*nearly ENP frame* if

$$(1 - \varepsilon)I_d \preceq X^T X \preceq (1 + \varepsilon)I_d, \quad (1 - \varepsilon)a \leq \|x_i\|^2 \leq (1 + \varepsilon)a \quad (1 \leq i \leq n). \quad (1.1)$$

Theorem 1.1 (Linear Paulsen bound). *There is an absolute constant C such that for all $1 \leq d \leq n$, all $0 < \varepsilon < 1$ and every real ε -nearly ENP frame $X \in \mathbb{R}^{n \times d}$ there is an ENP frame $W \in \mathbb{R}^{n \times d}$ with $\|X - W\|_F^2 \leq C\varepsilon d$.*

The bound is optimal up to the constant. Kwok, Lau, Lee and Ramachandran [3] proved the first bound independent of n , $O(\varepsilon d^{13/2})$; Hamilton and Moitra [2] improved it to $O(\varepsilon d^2)$; Lau and Ramachandran announced the linear bound [4]. The present text reorganises the proof of [5]; [Appendix B](#) lists what was changed.

For a Parseval U write $P = UU^T$ for the orthogonal projection onto its column space; then $P_{ii} = \|u_i\|^2$. The problem has an equivalent form in which the error is additive.

Theorem 1.2 (Projection form). *There is an absolute constant C such that for every $n \geq 1$ and every rank- d orthogonal projection P on $\mathbb{R}^n$, with $\beta = \max_i |P_{ii} - d/n|$, there is a rank- d orthogonal projection Q with $Q_{ii} = d/n$ for all i and $\|P - Q\|_F^2 \leq Cn\beta$.*

Both β and $\|P - Q\|_F$ are unchanged when (P, Q) is replaced by $(I - P, I - Q)$, so in [Theorem 1.2](#) one may assume $d \leq n/2$. The two theorems are derived from each other and from the main proposition in [Section 6](#).

Conventions. c, C denote positive absolute constants whose value may change between occurrences; constants with subscripts or names are fixed once chosen. $\|\cdot\|_{\text{op}}$ is the Euclidean operator norm, $\mathbf{1}$ the all-ones vector, $\Pi = I - n^{-1}\mathbf{1}\mathbf{1}^T$ the projection onto $\mathbf{1}^\perp$, $\text{Diag}(x)$ the diagonal matrix with diagonal x , and $\text{diag}(M)$ the diagonal of M as a vector. $A \preceq B$ is the Loewner order. A Laplacian inequality “on $\mathbf{1}^\perp$ ” means the corresponding quadratic forms are compared on vectors orthogonal to $\mathbf{1}$. All Gaussian vectors are real and centred.

1.2 The plan

The engine: scaling. Let U be Parseval with projection P and diagonal $p = \text{diag } P$. For $s \in \mathbb{R}^n$ the matrix $\text{Diag}(e^s)U$ has a column space with projection $P(s)$, and $s \mapsto \text{diag } P(s)$ is the gradient of the convex function $\frac{1}{2} \log \det(U^T \text{Diag}(e^{2s})U)$, whose Hessian at $s = 0$ is $2L_P$, where

$$L_P = \text{Diag}(p) - P \circ P$$

is the graph Laplacian with edge weights P_{ij}^2 (Lemma 2.2). To first order, then, changing the diagonal by $\delta \perp \mathbf{1}$ requires $s \approx \frac{1}{2}L_P^+ \delta$, and the cost of the move is about $s^T L_P s \approx \frac{1}{4} \delta^T L_P^+ \delta$. If L_P has spectral gap λ and $|\delta_i| \leq \varepsilon a$, the cost is at most $n\varepsilon^2 a^2 / \lambda$. For a “random-looking” P one has $\lambda \approx a$ and the cost is $\varepsilon^2 d$, but L_P can be arbitrarily badly connected: for an exact direct sum the components cannot exchange diagonal mass at all.

Perturb, then scale. Perturb U by a random horizontal tangent of size $t = \sqrt{K_0 \varepsilon}$, at cost $O(t^2 d) = O(\varepsilon d)$. Generic entries of the new projection then have size $t\sqrt{a/n}$, so most edges acquire weight $\gtrsim at^2/n$, and the random walk with rates P_{ij}^2 reaches any half of the vertices in time $O(1/(at^2))$: the *barrier constant* (Definition 2.6 and Lemma 2.10) is $O(1/(at^2))$. The static balancing theorem (Theorem 2.8) then removes any diagonal error β with $\beta \cdot O(1/(at^2)) \leq \frac{1}{2}$ exactly, at cost $\frac{15}{4}n\beta = O(t^2 d)$. The ℓ_∞ (rather than spectral) formulation is essential: the scaling must stay in a box $\|s\|_\infty \leq 1$, and an ℓ_2 argument loses a factor $\sqrt{n}$.

The difficulty. So the perturbed frame must have diagonal error $\leq \theta at^2$ in ℓ_∞ for a small fixed θ . The initial error $\varepsilon a = at^2/K_0$ is fine. But a random tangent of size t moves each diagonal entry *linearly* by about $t\sqrt{a/n}$, which exceeds at^2 as soon as $\varepsilon \lesssim 1/d$. Two different constructions avoid this.

Seed 1: many rows ($n \geq Bd^2$, Section 4). Perturb each row *orthogonally to itself* and renormalise. Row norms stay exact; the cost is that the frame operator is no longer exactly I . Its first order change is removed by conditioning the Gaussian on a subspace of codimension $\leq d^2$, and its second-order fluctuation is an average over n rows, of size $t^2 d / \sqrt{n}$. The spectral error of the perturbed frame is therefore $\eta + O(t^2 d / \sqrt{n} + t^2 d^2 / n + t^3)$, with t^2 a large multiple of η , which is at most θt^2 once that multiple and B are large and t is small; whitening turns a spectral error δ into a diagonal error at most $2a\delta$. (In [5] the remainder was bounded by $O(t^3 d)$, which forced $\eta \leq c/d^2$ and a separate block-partition argument; the sharper bound $O(t^3)$ uses the first-order constraint once more, see Lemma 4.4.)

Seed 2: moderate rows ($d \geq A \log 2n$, Section 5). Perturb in the horizontal tangent space of the Grassmannian (so the frame stays exactly Parseval) with a Gaussian whose covariance is filtered: $C_\rho = \rho(I - B)(\rho I + B)^{-1}$, where B is the normalised Gram operator of the linear diagonal

map $\mathcal{A}$. The filter makes the linear diagonal change $O(t\sqrt{\rho a \log n/n})$, negligible once $\rho \asymp t^2$ and $d \gg \log n$, while leaving weak (connecting) directions untouched. The price is a second-order bias $t^2 b_*$, which is not small in ℓ_∞ . A commutator identity shows that $b_* = \mathcal{A} m_*$ for an explicit deterministic tangent direction m_* with $\|m_*\|_F^2 \leq 2a/\rho$ (Lemmas 5.5 and 5.6). We simply add the drift $\frac{t}{2} m_*$ to the noise: its first-order effect cancels the bias exactly, and its other effects are $O(at^2/D)$. All remaining errors are $\leq \theta at^2$ by concentration, which is where $d \geq A \log 2n$ is used. Only a bounded, deterministic set of rows needs a spectral argument, and second moments suffice there.

Assembly (Section 6). After complementation and normalisation it suffices to treat equal-norm frames with $2d \leq n$ and spectral error η . If $d < D$ (a large absolute constant), the Hamilton–Moitra bound $\eta d(d-1) \leq D\eta d$ suffices (Section 3). If $d \geq D$ and $n \geq Bd^2$, use Seed 1. If $d \geq D$ and $n < Bd^2$, then $A \log 2n \leq A \log(2Bd^2) \leq d$ and Seed 2 applies.

regime	construction	where
$d < D$	radial isotropic position, ordered coordinates (Hamilton–Moitra)	Section 3
$d \geq D, n \geq Bd^2$	row-tangent Gaussian, row normalisation, whitening, balancing	Section 4
$d \geq D, 2d \leq n < Bd^2$	filtered horizontal Gaussian, bias correction, balancing	Section 5
$d > n/2$	complement	Section 6

2 The scaling toolbox

Throughout this section $U \in \mathbb{R}^{n \times d}$ is Parseval, $P = UU^T$, $p = \text{diag } P$, and $L = L_P = \text{Diag}(p) - P \circ P$. For nonnegative symmetric weights w_{ij} the Laplacian is $(Lx)_i = \sum_j w_{ij}(x_i - x_j)$; diagonal weights do not contribute. We write $\text{osc}(x) = \max_i x_i - \min_i x_i$.

2.1 Projections, distances, and the squared-Gram Laplacian

Lemma 2.1 (Frames versus projections). *Let $U, W \in \mathbb{R}^{n \times d}$ be Parseval with projections P, Q , and let $\sigma_1, \dots, \sigma_d \in [0, 1]$ be the singular values of $U^T W$. Then*

$$\min_{R \in O(d)} \|U - WR\|_F^2 = 2 \sum_k (1 - \sigma_k) \leq \|P - Q\|_F^2 = 2 \sum_k (1 - \sigma_k^2) = 2\|(I - P)W\|_F^2 \leq 2\|U - W\|_F^2.$$

Right multiplication by $R \in O(d)$ preserves Parsevalness and every row norm. If $X \in \mathbb{R}^{n \times d}$ has $\delta = \|X^T X - I\|_{\text{op}} < 1$, its polar factor $\tilde{X} = X(X^T X)^{-1/2}$ is Parseval, $\|X - \tilde{X}\|_F^2 \leq d\delta^2$, and $\|x_i\|^2/(1 + \delta) \leq \|\tilde{x}_i\|^2 \leq \|x_i\|^2/(1 - \delta)$.

Proof. $\|U - WR\|_F^2 = 2d - 2\text{tr}(U^T WR)$, maximised over R by the polar factor of $U^T W$, giving $2d - 2\sum_k \sigma_k$. Next $\|P - Q\|_F^2 = 2d - 2\text{tr}(PQ) = 2d - 2\|U^T W\|_F^2$ and $\|(I - P)W\|_F^2 = d - \|U^T W\|_F^2$; and $1 - \sigma \leq 1 - \sigma^2 \leq 2(1 - \sigma)$ on $[0, 1]$. For the polar factor, if ς_k are the singular values of X then $\|X - \tilde{X}\|_F^2 = \sum_k (\varsigma_k - 1)^2 \leq \sum_k (\varsigma_k^2 - 1)^2$, and $\|\tilde{x}_i\|^2 = x_i^T (X^T X)^{-1} x_i$. $\square$

Lemma 2.2 (Energy identity). *L is the Laplacian of the weights P_{ij}^2 ($i \neq j$), $L \succeq 0$, $L\mathbf{1} = 0$, $L_{I-P} = L_P$, and for $x \in \mathbb{R}^n$, $X = \text{Diag}(x)$,*

$$x^T L x = \frac{1}{2} \sum_{i,j} P_{ij}^2 (x_i - x_j)^2 = \|(I - P)XU\|_F^2 = \frac{1}{2} \|[X, P]\|_F^2.$$

Proof. $\sum_j P_{ij}^2 = (P^2)_{ii} = p_i$, so the rows of $\text{Diag}(p) - P \circ P$ sum to zero and its off-diagonal entries are $-P_{ij}^2$. Next $\|(I - P)XU\|_F^2 = \text{tr}(XPX(I - P)) = \sum_i x_i^2 p_i - \sum_{i,j} x_i x_j P_{ij}^2$, and $[X, P]_{ij} = (x_i - x_j)P_{ij}$. For $I - P$ the off-diagonal weights are again P_{ij}^2 . $\square$

Lemma 2.3 (Scaled projections). *For $w \in \mathbb{R}_{>0}^n$ put $D = \text{Diag}(w)$, $M = (U^T D^2 U)^{-1/2}$ and $W = DUM$. Then W is Parseval with column space $D \text{im } U$, and*

$$\|WW^T - P\|_F^2 = 2\|(I - P)W\|_F^2 \leq \frac{2w^T Lw}{\min_i w_i^2}.$$

Proof. $W^T W = MU^T D^2 U M = I$ and $U^T D^2 U \succeq (\min_i w_i)^2 I$, so $\|M\|_{\text{op}} \leq 1/\min_i w_i$. By Lemmas 2.1 and 2.2, $\|(I - P)W\|_F^2 \leq \|M\|_{\text{op}}^2 \|(I - P)DU\|_F^2 = \|M\|_{\text{op}}^2 w^T Lw$. $\square$

2.2 Diagonal scaling

For $s \in \mathbb{R}^n$ let $D_s = \text{Diag}(e^s)$, let $P(s)$ be the projection onto $D_s \text{im } U$, $r(s) = \text{diag } P(s)$, and $F(s) = \frac{1}{2} \log \det(U^T D_s^2 U)$.

Lemma 2.4. *F is smooth and convex, $\nabla F(s) = r(s)$, $\sum_i r_i(s) = d$, and $F(s + c\mathbf{1}) = F(s) + cd$.*

Proof. With $G = U^T D_s^2 U = \sum_i e^{2s_i} u_i u_i^T$ one has $\partial_i F = e^{2s_i} u_i^T G^{-1} u_i = P(s)_{ii}$, because $P(s) = D_s U G^{-1} U^T D_s$; differentiating again gives $\nabla^2 F = 2L_{P(s)} \succeq 0$. The trace of $P(s)$ is d , and G scales by e^{2c} under $s \mapsto s + c\mathbf{1}$. $\square$

Lemma 2.5 (One-sided scaling inequalities). *For $z \in \mathbb{R}_{>0}^n$ let r be the diagonal of the projection onto $\text{Diag}(\sqrt{z}) \text{im } U$. Then, coordinatewise,*

$$Lz \leq z \circ (r - p), \quad L(z^{-1}) \leq -z^{-1} \circ (r - p).$$

Proof. Let $G = U^T \text{Diag}(z)U$. Since $G + z_i^2 G^{-1} - 2z_i I = (G^{1/2} - z_i G^{-1/2})^2 \succeq 0$, evaluating at u_i and using $u_i^T G u_i = \sum_j P_{ij}^2 z_j = p_i z_i - (Lz)_i$ and $z_i^2 u_i^T G^{-1} u_i = z_i r_i$ gives the first inequality. Let $V \in \mathbb{R}^{n \times (n-d)}$ be an isometry onto $(\text{im } U)^\perp$. The subspaces $\text{Diag}(\sqrt{z}) \text{im } U$ and $\text{Diag}(z^{-1/2}) \text{im } V$ are orthogonal complements, so the projection onto the latter has diagonal $\mathbf{1} - r$, while $VV^T = I - P$ has diagonal $\mathbf{1} - p$ and Laplacian L (Lemma 2.2). The first inequality for (V, z^{-1}) is the second. $\square$

2.3 The barrier constant and static balancing

Definition 2.6 (Barrier constant). Let L be a weighted Laplacian on $[n]$. For $S \subseteq [n]$, an S -barrier is a vector $\psi \geq 0$ with $(L\psi)_i \geq 1$ for every $i \notin S$. The barrier constant $H(L)$ is the least H such that every S with $|S| \geq n/2$ has an S -barrier with $\|\psi\|_\infty \leq H$.

The minimal S -barrier is $x \mapsto \mathbb{E}_x \tau_S$, the expected hitting time of S for the continuous-time random walk that jumps from i to j at rate w_{ij} ; so $H(L_P)$ is the half-set hitting time of the walk with rates P_{ij}^2 . It is within a factor 2 of the ℓ_∞ Poisson constant used in [5] (Remark 2.11). A small barrier constant controls scalings in ℓ_∞ :

Lemma 2.7 (Median barrier). *Let L have barrier constant H , let $\beta \geq 0$ with $\beta H < 1$, let $z \in \mathbb{R}_{>0}^n$, and let m be a median of z (so that $\{z \leq m\}$ and $\{z \geq m\}$ both have at least $n/2$ elements). If*

$$(Lz)_i \leq \beta z_i \quad \text{when } z_i > m, \quad (Lz^{-1})_i \leq \beta z_i^{-1} \quad \text{when } z_i < m,$$

then $(1 - \beta H) \max_i z_i \leq m \leq (1 - \beta H)^{-1} \min_i z_i$.

Proof. Let $S = \{z \leq m\}$, let ψ be an S -barrier with $\|\psi\|_\infty \leq H$, let $M = \max z$ and $\beta' > \beta$. The vector $v = z - m\mathbf{1} - \beta'M\psi$ is ≤ 0 on S , and for $i \notin S$, $(Lv)_i \leq \beta z_i - \beta'M < 0$. At a maximiser i of v , $(Lv)_i = \sum_j w_{ij}(v_i - v_j) \geq 0$; so every maximiser lies in S , and $v \leq 0$. Hence $M \leq m + \beta'MH$; let $\beta' \downarrow \beta$. Apply the same argument to z^{-1} , whose median is m^{-1} , with $S = \{z \geq m\}$. $\square$

Theorem 2.8 (Static balancing). *Suppose L_P has barrier constant H , and $q \in \mathbb{R}^n$ satisfies $\sum_i q_i = d$ and $\beta H \leq \frac{1}{2}$, where $\beta = \|q - p\|_\infty$. Then there is a Parseval W with $\text{diag}(WW^T) = q$ and*

$$\|U - W\|_F^2 \leq \|UU^T - WW^T\|_F^2 \leq \frac{15}{4}\|q - p\|_1.$$

W is obtained from U by a row scaling with ratio of scales at most 2, whitening, and an orthogonal right factor.

Proof. Let s minimise $\Phi(s) = F(s) - \langle q, s \rangle$ over the cube $[0, 1]^n$ (compactness), and write $r = r(s)$. Since $\partial_i \Phi = r_i - q_i$ (Lemma 2.4), coordinatewise optimality gives

$$r_i \leq q_i \text{ if } s_i > 0, \quad r_i \geq q_i \text{ if } s_i < 1. \quad (2.1)$$

Let $z = e^{2s}$ with median m , so $1 \leq \min z \leq m \leq \max z \leq e^2$. If $z_i > m$ then $s_i > 0$, so by Lemma 2.5 and (2.1) $(Lz)_i \leq z_i(r_i - p_i) \leq z_i(q_i - p_i) \leq \beta z_i$; if $z_i < m$ then $s_i < 1$ and likewise $(Lz^{-1})_i \leq \beta z_i^{-1}$. By Lemma 2.7, $\max z / \min z \leq (1 - \beta H)^{-2} \leq 4$, i.e. $\text{osc}(s) \leq \log 2 < 1$. Hence s does not meet both faces of the cube: either all $s_i > 0$, and then $r \leq q$, or some $s_j = 0$, all $s_i < 1$, and then $r \geq q$. Since $\sum r_i = d = \sum q_i$, in both cases $r = q$.

For the distance let $w = e^s$, $f = q - p$, and let c be the midpoint of the range of z^2 . Lemma 2.5 with $r = q$ gives $Lz \leq z \circ f$, so, using $\sum_i f_i = 0$,

$$z^T Lz \leq \sum_i z_i^2 f_i = \sum_i (z_i^2 - c) f_i \leq \frac{1}{2} \text{osc}(z^2) \|f\|_1.$$

As $(z_i - z_j)^2 \geq 4 \min w^2 (w_i - w_j)^2$, Lemmas 2.2 and 2.3 give

$$\|P(s) - P\|_F^2 \leq \frac{2w^T Lw}{\min w^2} \leq \frac{z^T Lz}{2 \min z^2} \leq \frac{\text{osc}(z^2)}{4 \min z^2} \|f\|_1 \leq \frac{4^2 - 1}{4} \|f\|_1.$$

Finally apply Lemma 2.1 to the whitened frame $DU(U^T D^2 U)^{-1/2}$, $D = \text{Diag}(w)$, whose projection is $P(s)$. $\square$

Remark 2.9. No existence theorem for scalings, no flow, no genericity and no connectivity argument is used: the box supplies a minimiser, the median barrier keeps it away from one face, and the trace identity $\sum_i r_i = d$ does the rest. With $\beta H < 1$ and a box of side $T > -\log(1 - \beta H)$ the same proof gives scale ratio $(1 - \beta H)^{-1}$ and cost $\frac{1}{4}((1 - \beta H)^{-4} - 1)\|q - p\|_1$.

2.4 Barriers from a dense core

Lemma 2.10 (Dense core with exceptional vertices). *Let L be the Laplacian of nonnegative symmetric weights w_{ij} on $[n]$, and let $[n] = G \sqcup B$. Suppose:*

- (a) *there is $\kappa > 0$ with $\sum_{j \in S} w_{ij} \geq \kappa$ for every $i \in G$ and every $S \subseteq [n] \setminus \{i\}$ with $|S| \geq n/2$;*
- (b) *there are $y \in \mathbb{R}^B$, $y \geq 0$, and $R, F \geq 0$ with $\|y\|_\infty \leq R$, $(L_B B y)_i \geq 1$ for $i \in B$, and $\sum_{j \in B} w_{ij} y_j \leq F$ for $i \in G$. (If $B = \emptyset$ take $R = F = 0$.)*

Then $H(L) \leq (1 + F)/\kappa + R$. Moreover:

- (i) (a) holds with $\kappa = \vartheta\gamma$ if every $i \in G$ has at least $(\frac{1}{2} + \vartheta)n$ indices $j \neq i$ with $w_{ij} \geq \gamma/n$;
- (ii) (b) holds with $R = \sqrt{b}/\lambda$, $F = b$, where $b = |B|$, if $x^T Lx \geq \lambda\|x\|^2$ for every x supported on B ($\lambda > 0$);
- (iii) for $w_{ij} = P_{ij}^2$, (b) holds with $R = \frac{8}{3a}$, $F = \frac{2}{3a} \max_i p_i$ if $a > 0$, $p_i \geq a/2$ on B and $\sum_{i \in B} p_i \leq \frac{1}{4}$.

Proof. Let $|S| \geq n/2$, put $\lambda_0 = (1 + F)/\kappa$, extend y by 0 to G , and let $\psi = (\lambda_0 \mathbf{1} + y) \circ \mathbf{1}_{S^c}$, so $0 \leq \psi \leq \lambda_0 + R$ and $\psi_j \leq \lambda_0 + y_j$ for all j . For $i \in G \setminus S$, $\psi_i = \lambda_0$ and $\psi_j = 0$ on S , so

$$(L\psi)_i \geq \lambda_0 \sum_{j \in S} w_{ij} - \sum_{j \in B} w_{ij} y_j \geq \lambda_0 \kappa - F = 1.$$

For $i \in B \setminus S$, $\psi_i = \lambda_0 + y_i$, so $(L\psi)_i \geq \sum_j w_{ij} (y_i - y_j) = (L_{BB} y)_i \geq 1$. Thus ψ is an S -barrier.

(i) If N_i is the set of such j and $S \subseteq [n] \setminus \{i\}$, then $|N_i \cap S| \geq |N_i| + |S| - (n - 1) > \vartheta n$. (ii) $C = L_{BB} \succeq \lambda I$; let $y = C^{-1} \mathbf{1}_B$. If y had a negative minimum at i , then $(Cy)_i = y_i \sum_{j \notin B} w_{ij} + \sum_{j \in B} w_{ij} (y_i - y_j) \leq 0$, a contradiction; so $y \geq 0$, and $\|y\|_\infty \leq \|y\|_2 \leq \sqrt{b}/\lambda$. Since $(C \mathbf{1}_B)_j = \sum_{i \in G} w_{ij}$ for $j \in B$, for each $i_0 \in G$

$$\sum_{j \in B} w_{i_0 j} y_j \leq \sum_{i \in G} \sum_{j \in B} w_{ij} y_j = \mathbf{1}_B^T C y = b.$$

(iii) $P_{ij}^2 \leq p_i p_j$ gives, for $i \in B$, $(L_{BB} \mathbf{1}_B)_i = p_i - \sum_{j \in B} P_{ij}^2 \geq p_i (1 - \sum_{j \in B} p_j) \geq \frac{3a}{8}$, and for $i \in G$, $\sum_{j \in B} P_{ij}^2 \leq p_i/4$. Take $y = \frac{8}{3a} \mathbf{1}_B$. $\square$

Remark 2.11 (Poisson constants). [5] uses the Poisson constant K : every $f \perp \mathbf{1}$ has a solution of $Lg = f$ with $\|g\|_\infty \leq K\|f\|_\infty$. Then $K \leq H \leq 2K$. Indeed, for $|S| \geq n/2$ the forcing $f = \mathbf{1} - \frac{n}{|S|} \mathbf{1}_S$ has $\|f\|_\infty \leq 1$, and $g + K\mathbf{1}$ is an S -barrier; conversely, if $Lg = f$ with $\|f\|_\infty \leq 1$ and m is a median of g , the argument of Lemma 2.7 applied to $g - m\mathbf{1} - \beta'\psi$ ($\beta' > 1$) gives $g \leq m + H$, and symmetrically $g \geq m - H$. Lemma 2.10 replaces the Schur-complement estimate of [5] by an explicit barrier.

2.5 From a seed to an exact frame

Both constructions produce the following object.

Definition 2.12 (Seed). Let $\Theta \geq 2$ and $t > 0$. A (Θ, t) -seed for $X \in \mathbb{R}^{n \times d}$ is a Parseval $V \in \mathbb{R}^{n \times d}$ with

$$\|V - X\|_F^2 \leq \Theta t^2 d, \quad H(L_{VV^T}) \leq \frac{\Theta}{at^2}, \quad \max_i |(VV^T)_{ii} - a| \leq \frac{at^2}{2\Theta}.$$

Corollary 2.13. If X has a (Θ, t) -seed, there is an ENP frame W with $\|X - W\|_F^2 \leq 3\Theta t^2 d$.

Proof. Apply Theorem 2.8 to V with $q = a\mathbf{1}$: $\beta H \leq \frac{1}{2}$, and $\|V - W\|_F^2 \leq \frac{15}{4}n \cdot \frac{at^2}{2\Theta} \leq t^2 d$. Then $\|X - W\|_F^2 \leq 2\|X - V\|_F^2 + 2\|V - W\|_F^2 \leq (2\Theta + 2)t^2 d \leq 3\Theta t^2 d$. $\square$

3 Bounded rank: the Hamilton–Moitra estimate

This section reproduces the radial-isotropic and ordered-coordinate argument of Hamilton and Moitra [2], with a direct compactness proof of the existence of the radial scaling (as in [5]). It is used for d below a large absolute constant, and it also shows that ENP frames exist.

Lemma 3.1. *Let $1 \leq d \leq n$, let $X \in \mathbb{R}^{n \times d}$ have $\|x_i\|^2 = a$ for all i , and let $\eta = \|X^T X - I\|_{\text{op}}$. There is an ENP frame W with $\|X - W\|_{\text{F}}^2 \leq \eta d(d-1)$.*

Proof. If $d = 1$ then $X^T X = na = 1$ and $W = X$ works. Let $d \geq 2$.

Step 1: full spark. Suppose first that every $d \times d$ minor $\mu_S = \det(X_S)^2$ ($S \subset [n]$, $|S| = d$) is nonzero. Consider $F(s) = \frac{1}{2} \log \det(X^T \text{Diag}(e^{2s})X) - a \sum_i s_i$. By Cauchy–Binet,

$$F(s) = \frac{1}{2} \log \sum_{|S|=d} \mu_S e^{T(s)S}, \quad T(s)_S = 2 \left(\sum_{i \in S} s_i - a \sum_i s_i \right),$$

where $T : \mathbb{R}^n \rightarrow \mathbb{R}^{(n)}_d$ is linear and $\sum_S T(s)_S = 0$ because each i lies in a fraction $d/n = a$ of the d -sets. The function $\mathcal{E}(y) = \sum_S \mu_S e^{y_S}$ restricted to the subspace $\text{im } T$ has the nonempty sublevel set $\{\mathcal{E} \leq \mathcal{E}(0)\}$, on which $y_S \leq \log(\mathcal{E}(0)/\mu_S)$ for every S and hence, by $\sum_S y_S = 0$, also y_S is bounded below. This set is compact, so $\mathcal{E}|_{\text{im } T}$ attains its minimum at some $T(s^*)$, and s^* minimises F .

Let $D = \text{Diag}(e^{s^*})$, $M = (X^T D^2 X)^{-1/2}$ and $W = DXM$. As in Lemma 2.4, $\partial_i F = \|w_i\|^2 - a$, so W is ENP, and since $w_i = e^{s_i^*} Mx_i$ has norm $\sqrt{a}$,

$$w_i = \sqrt{a} \frac{Mx_i}{\|Mx_i\|}.$$

Step 2: ordered coordinates. Replacing X, W by XO, WO for a suitable $O \in O(d)$ changes neither the hypotheses nor the distance, and makes $M = \text{Diag}(\lambda_1, \dots, \lambda_d)$ with $0 < \lambda_1 \leq \dots \leq \lambda_d$. Then $w_{ij} = \lambda_j x_{ij} / \sqrt{c_i}$ with $c_i = a^{-1} \sum_\ell \lambda_\ell^2 x_{i\ell}^2$, so w_{ij} has the sign of x_{ij} and $w_{ij}^2 - x_{ij}^2 = x_{ij}^2 (\lambda_j^2 / c_i - 1)$ is nonpositive and then nonnegative as j increases. As $\sum_j (x_{ij}^2 - w_{ij}^2) = a - a = 0$, the prefix sums $\Delta_i(k) = \sum_{j \leq k} (x_{ij}^2 - w_{ij}^2)$ are nonnegative, vanish at $k = d$, and

$$\sum_j |x_{ij}^2 - w_{ij}^2| = 2 \max_k \Delta_i(k) \leq 2 \sum_{k=1}^{d-1} \Delta_i(k).$$

Since $W^T W = I$ and $(X^T X)_{jj} \leq 1 + \eta$, $\sum_i \Delta_i(k) = \sum_{j \leq k} ((X^T X)_{jj} - 1) \leq k\eta$. Equal signs give $(x_{ij} - w_{ij})^2 \leq |x_{ij}^2 - w_{ij}^2|$, and therefore

$$\|X - W\|_{\text{F}}^2 \leq 2\eta \sum_{k=1}^{d-1} k = \eta d(d-1).$$

Step 3: general X . Let G be a Vandermonde matrix with rows $(1, \theta_i, \dots, \theta_i^{d-1})$ for distinct θ_i ; all its maximal minors are nonzero. Each maximal minor of $X + hG$ is a polynomial in h with nonzero leading coefficient, so for all but finitely many h the matrix $X + hG$ is full spark; choose such $h_k \rightarrow 0$ and normalise each row of $X + h_k G$ to length $\sqrt{a}$ (a positive diagonal left factor, which preserves full spark). The resulting $X_k \rightarrow X$ have ENP frames W_k with $\|X_k - W_k\|_{\text{F}}^2 \leq \eta_k d(d-1)$, $\eta_k \rightarrow \eta$. The set of ENP frames is compact; a limit point W of (W_k) is ENP and satisfies the bound. $\square$

Corollary 3.2. *For all $1 \leq d \leq n$ an ENP frame exists. Consequently every equal-norm $X \in \mathbb{R}^{n \times d}$ is within $\|X - W\|_F^2 \leq 2\|X\|_F^2 + 2\|W\|_F^2 = 4d$ of an ENP frame W .*

Proof. Apply Lemma 3.1 to $x_i = \sqrt{a} e_1$ for all i . □

4 The many-row seed

Theorem 4.1. *There are absolute constants $B, c_*, C_* > 0$ such that the following holds. Let $2 \leq d, n \geq Bd^2$, let $X \in \mathbb{R}^{n \times d}$ have $\|x_i\|^2 = a$ for all i , and suppose $\eta := \|X^T X - I\|_{\text{op}}$ satisfies $0 < \eta \leq c_*$. Then there is an ENP frame W with $\|X - W\|_F^2 \leq C_* \eta d$.*

In [5] the same construction is analysed under the stronger hypothesis $\eta \leq c/d^2$; the improvement comes from Lemma 4.4. Throughout, $S = X^T X$, so $\|S - I\|_{\text{op}} \leq \eta \leq \frac{1}{2}$, and $e_i = x_i/\sqrt{a}$.

4.1 The noise

In the Frobenius space $\mathbb{R}^{n \times d}$ consider the subspaces

$$\mathcal{E} = \{Z : \langle x_i, z_i \rangle = 0 \ \forall i\}, \quad \mathcal{F} = \{Z \in \mathcal{E} : X^T Z + Z^T X = 0\},$$

and let $m = \dim \mathcal{E} - \dim \mathcal{F} \leq \dim \text{Sym}_d \leq d^2$. For a standard Gaussian $g \in \mathbb{R}^{n \times d}$ put

$$Z_0 = n^{-1/2} \Pi_{\mathcal{E}} g, \quad Z = n^{-1/2} \Pi_{\mathcal{F}} g, \quad Z_{\perp} = Z_0 - Z = n^{-1/2} \Pi_{\mathcal{E} \ominus \mathcal{F}} g.$$

Z and $Z_{\perp}$ are independent. The rows $z_{0,i} = n^{-1/2}(I - e_i e_i^T)g_i$ of Z_0 are independent. Since $\text{Cov}(Z) \preceq \text{Cov}(Z_0)$, each row z_i of Z is Gaussian with covariance $\Sigma_i \preceq n^{-1}(I - e_i e_i^T)$, and $\text{Cov}(\text{vec } Z) \preceq I/n$. Moreover

$$\mathbb{E}\|Z\|_F^2 = \frac{n(d-1) - m}{n} \leq d, \quad \mathbb{E}\|Z - Z_0\|_F^2 = \frac{m}{n} \leq \frac{d^2}{n}. \quad (4.1)$$

Lemma 4.2 (Row moments). *Let $r_i = \|z_i\|^2/a$, $\mu_i = \mathbb{E}r_i$ and $\delta_i = 1 - \mu_i$. Then $1/d \leq \delta_i \leq 1$, $\sum_i a\delta_i = 1 + m/n \leq 2$, and*

$$\mathcal{A} := a \sum_i \mathbb{E}[(r_i - 1)^2(1 + r_i)^2] \leq C_{\mathcal{A}},$$

with $C_{\mathcal{A}}$ absolute.

Proof. $\tilde{\Sigma}_i = \Sigma_i/a \preceq d^{-1}I$ has trace $\mu_i \leq (d-1)/d$, so $\text{tr } \tilde{\Sigma}_i^2 \leq 1/d$ and $\text{tr } \tilde{\Sigma}_i^4 \leq d^{-3}$. For $\xi_i = r_i - \mu_i$ the Gaussian moment formulas give $\mathbb{E}\xi_i^2 = 2 \text{tr } \tilde{\Sigma}_i^2 \leq 2/d$ and $\mathbb{E}\xi_i^4 = 12(\text{tr } \tilde{\Sigma}_i^2)^2 + 48 \text{tr } \tilde{\Sigma}_i^4 \leq 60/d^2$. Next $\sum_i a\delta_i = d - \mathbb{E}\|Z\|_F^2 = 1 + m/n$ by (4.1). Finally $(r_i - 1)^2 \leq 2\xi_i^2 + 2\delta_i^2$ and $(1 + r_i)^2 \leq 8 + 2\xi_i^2$, so

$$\mathbb{E}(r_i - 1)^2(1 + r_i)^2 \leq 16\mathbb{E}\xi_i^2 + 4\mathbb{E}\xi_i^4 + 16\delta_i^2 + 4\delta_i^2\mathbb{E}\xi_i^2 \leq \frac{C}{d} + C\delta_i^2,$$

and $a \sum_i (C/d + C\delta_i^2) \leq C + Ca \sum_i \delta_i \leq 3C$. □

4.2 Row normalisation and the exact second-order identity

For $0 < t \leq 1$ set

$$v_i = \frac{x_i + tz_i}{\sqrt{1 + t^2 r_i}}, \quad v_{0,i} = \frac{x_i + tz_{0,i}}{\sqrt{1 + t^2 r_{0,i}}}, \quad r_{0,i} = \|z_{0,i}\|^2/a.$$

Since $z_i \perp x_i$, $\|v_i\|^2 = \|v_{0,i}\|^2 = a$ exactly. The map $y \mapsto \sqrt{a}y/\|y\|$ is the metric projection onto the closed ball of radius $\sqrt{a}$ on $\{\|y\| \geq \sqrt{a}\}$, hence 1-Lipschitz there. Thus

$$\|V - X\|_F \leq t\|Z\|_F, \quad \|V - V_0\|_F \leq t\|Z - Z_0\|_F, \quad \|V\|_{\text{op}} \leq \|X + tZ\|_{\text{op}}, \quad \|V_0\|_{\text{op}} \leq \|X + tZ_0\|_{\text{op}}, \quad (4.2)$$

the last two because $V = \text{Diag}(\kappa)(X + tZ)$ with $0 < \kappa_i \leq 1$.

Expanding $1/(1 + t^2 r_i) = 1 - t^2 c_i$ with

$$c_i = \frac{r_i}{1 + t^2 r_i},$$

and using $X^T Z + Z^T X = 0$, one finds the exact identity

$$V^T V = S + t^2 Q(Z) + R_3 + R_4, \quad Q(Z) = Z^T Z - \sum_i r_i x_i x_i^T, \quad (4.3)$$

$$R_3 = -t^3 \sum_i c_i (x_i z_i^T + z_i x_i^T), \quad R_4 = -t^4 \sum_i c_i (z_i z_i^T - r_i x_i x_i^T).$$

Lemma 4.3 (Mean and fluctuation of Q). $\|\mathbb{E}Q(Z) - (I - S)\|_{\text{op}} \leq m/n$ and $\mathbb{E}\|Q(Z) - \mathbb{E}Q(Z)\|_F^2 \leq 8d^2/n$.

Proof. For Z_0 : $\mathbb{E}Z_0^T Z_0 = n^{-1} \sum_i (I - e_i e_i^T) = I - S/d$ and $\mathbb{E}r_{0,i} = (d - 1)/d$, so $\mathbb{E}Q(Z_0) = I - S/d - (1 - 1/d)S = I - S$. As $Z_0 = Z + Z_\perp$ with independent centred summands and Q is quadratic, $\mathbb{E}Q(Z) = \mathbb{E}Q(Z_0) - \mathbb{E}Q(Z_\perp)$, and $\mathbb{E}Q(Z_\perp)$ is the difference of the positive semidefinite matrices $\mathbb{E}Z_\perp^T Z_\perp$ and $\sum_i \mathbb{E}(\|z_{\perp,i}\|^2/a) x_i x_i^T$, both of trace $\mathbb{E}\|Z_\perp\|_F^2 = m/n$; so its norm is at most m/n .

Entry (α, β) of Q is the quadratic form $\sum_i z_i^T B_{\alpha\beta,i} z_i$ with $B_{\alpha\beta,i} = \frac{1}{2}(E_{\alpha\beta} + E_{\beta\alpha}) - (e_i)_\alpha (e_i)_\beta I$. For a Gaussian vector with covariance $\Sigma \preceq I/n$ and symmetric coefficient matrix $\mathbf{B}$, the variance of the quadratic form is $2\|\Sigma^{1/2} \mathbf{B} \Sigma^{1/2}\|_F^2 \leq 2\|\mathbf{B}\|_F^2/n^2$. Since $\sum_{\alpha,\beta} \|B_{\alpha\beta,i}\|_F^2 \leq 2d^2 + 2d \leq 4d^2$ for each i , summing over α, β and the n blocks gives the claim. $\square$

Lemma 4.4 (Remainder). Let $\bar{c} = 1/(1 + t^2)$, $\Gamma = \text{Diag}(c_i - \bar{c})$ and $\Lambda = \text{Diag}(c_i r_i - \bar{c})$. Then

$$\|R_3\|_{\text{op}} \leq 2t^3 \|X\|_{\text{op}} \|\Gamma Z\|_F, \quad \|R_4\|_{\text{op}} \leq t^4 (\|Z\|_{\text{op}}^2 + \|Z\|_{\text{op}} \|\Gamma Z\|_F + \|S\|_{\text{op}} + \|X\|_{\text{op}} \|\Lambda X\|_F),$$

and $\mathbb{E}\|\Gamma Z\|_F^2 \leq \mathcal{A}$, $\mathbb{E}\|\Lambda X\|_F^2 \leq 4\mathcal{A}$.

Proof. Because $\sum_i (x_i z_i^T + z_i x_i^T) = X^T Z + Z^T X = 0$, the constant part $\bar{c}$ of the weights drops out of R_3 : $R_3 = -t^3 (X^T \Gamma Z + Z^T \Gamma X)$, which gives the first bound. Next $\sum_i c_i z_i z_i^T = \bar{c} Z^T Z + Z^T \Gamma Z$ and $\sum_i c_i r_i x_i x_i^T = \bar{c} S + X^T \Lambda X$, with $\|Z^T \Gamma Z\|_{\text{op}} \leq \|Z\|_{\text{op}} \|\Gamma Z\|_F$ and similarly for Λ , and $\bar{c} \leq 1$. For the moments,

$$c_i - \bar{c} = \frac{r_i - 1}{(1 + t^2 r_i)(1 + t^2)}, \quad c_i r_i - \bar{c} = \frac{(r_i - 1)(1 + r_i + t^2 r_i)}{(1 + t^2 r_i)(1 + t^2)},$$

so $|c_i - \bar{c}| \leq |r_i - 1|$ and $|c_i r_i - \bar{c}| \leq 2|r_i - 1|(1 + r_i)$. Hence $\|\Gamma Z\|_F^2 = a \sum_i (c_i - \bar{c})^2 r_i \leq a \sum_i (r_i - 1)^2 (1 + r_i)^2$ and $\|\Lambda X\|_F^2 = a \sum_i (c_i r_i - \bar{c})^2 \leq 4a \sum_i (r_i - 1)^2 (1 + r_i)^2$; take expectations and use Lemma 4.2. $\square$

Remark 4.5. The crude bound $\|R_3\|_{\text{op}} \leq 2t^3 \sum_i c_i \|x_i\| \|z_i\| \approx 2t^3 d$ is what forces $\eta \lesssim d^{-2}$ in [5]. Centring the weights at $\bar{c}$, which is allowed by the first-order constraint, replaces the n terms of size a by fluctuations of relative size $d^{-1/2}$, whose ℓ_2 -sum over the rows is $O(1)$.

4.3 A dense reference graph

Lemma 4.6. *There are absolute $h, t_0 \in (0, 1]$ and $c_1 > 0$ such that for $t \leq t_0$, with probability at least $1 - ne^{-c_1(n-1)}$, every i has at least $0.9(n-1)$ indices $j \neq i$ with*

$$|\langle v_{0,i}, v_{0,j} \rangle| \geq h t \sqrt{a/n}.$$

Proof. Write $z_{0,j}/\sqrt{a} = d^{-1/2} \gamma_j$ with $\gamma_j \sim N(0, I - e_j e_j^T)$ independent. Fix i and condition on row i ; let $f = v_{0,i}/\sqrt{a}$, a unit vector. For $j \neq i$,

$$\frac{\langle v_{0,i}, v_{0,j} \rangle}{a} = \frac{\nu_j}{N_j}, \quad \nu_j = u_j + \frac{t}{\sqrt{d}} \langle f, \gamma_j \rangle, \quad u_j = \langle f, e_j \rangle, \quad N_j = (1 + \frac{t^2}{d} \|\gamma_j\|^2)^{1/2} \geq 1,$$

and the target inequality reads $|\nu_j|/N_j \geq ht/\sqrt{d}$. By Markov, $\mathbb{P}(N_j > 2) \leq t^2/3$. On $\{N_j \leq 2\}$ failure requires $|\nu_j| < 2ht/\sqrt{d}$. Here $\nu_j \sim N(u_j, t^2(1 - u_j^2)/d)$. If $|u_j| \leq \frac{1}{2}$ the variance is at least $3t^2/(4d)$ and the Gaussian density bound gives probability at most $4h/\sqrt{3\pi/2} \leq 3h$. If $|u_j| > \frac{1}{2}$ and $2ht \leq \frac{1}{4}$, failure requires $\frac{t}{\sqrt{d}} |\langle f, \gamma_j \rangle| > \frac{1}{4}$, of probability at most $16t^2/d$. Choose h and t_0 so that $3h + t_0^2/3 + 8t_0^2 \leq \frac{1}{50}$. The failure indicators for $j \neq i$ are conditionally independent with probabilities at most $\frac{1}{50}$, so by the exponential Markov inequality (with parameter 1) more than $0.1(n-1)$ failures have probability at most $\exp(-(n-1)[0.1 - \frac{e-1}{50}]) \leq e^{-c_1(n-1)}$. Take a union bound over i . $\square$

4.4 Whitening

Lemma 4.7. *Let V be equal-norm with $\delta = \|V^T V - I\|_{\text{op}} \leq \frac{1}{2}$, $U = V(V^T V)^{-1/2}$ and $P = UU^T$. Then $|P_{ii} - a| \leq 2a\delta$, $\sum_j (P - VV^T)_{ij}^2 \leq 6a\delta^2$ for each i , and $\|U - V\|_{\text{F}}^2 \leq d\delta^2$.*

Proof. $P_{ii} = v_i^T (V^T V)^{-1} v_i \in [a/(1+\delta), a/(1-\delta)]$. With $E = (V^T V)^{-1} - I$, $\|E\|_{\text{op}} \leq \delta/(1-\delta) \leq 2\delta$, row i of $P - VV^T = VE V^T$ has squared norm $v_i^T E V^T V E v_i \leq 4\delta^2(1+\delta)a$. Finally the singular values σ_k of V satisfy $|\sigma_k - 1| \leq |\sigma_k^2 - 1| \leq \delta$. $\square$

4.5 Proof of Theorem 4.1

Constants. Fix h, t_0, c_1 from Lemma 4.6. Put $\kappa = h^2/200$ and $C_{\text{mix}} = 125/h^2 + 3$, then $\Theta = \max\{42, C_{\text{mix}}\}$ and $\theta = \min\{1/(4\Theta), h/60, 1/16\}$. Let $C_R = 300 + 100\sqrt{C_{\mathcal{A}}}$, choose $t_1 = \min\{t_0, \theta/(16C_R)\}$, and set

$$t^2 = 4\eta/\theta, \quad c_* = \theta t_1^2/4,$$

so that $\eta \leq c_*$ gives $t \leq t_1$. Finally choose B so large that $n \geq Bd^2$ implies $d^2/n \leq \theta/4$, $\sqrt{160d^2/n} \leq \theta/4$, $\sum_{i \in B_0} 2a \leq \frac{1}{4}$ for the set B_0 below (which has $|B_0| \leq C'd/\kappa$), $n \geq 30$, and $ne^{-c_1(n-1)} \leq 0.01$.

The event. By Markov's inequality, (4.1), Lemmas 4.3 and 4.4, the net bound (Lemma A.1(c)) for Z and Z_0 , and Lemma 4.6, with positive probability all of the following hold:

$$\|Z\|_{\text{F}}^2 \leq 20d, \quad \|Z - Z_0\|_{\text{F}}^2 \leq \frac{20d^2}{n}, \quad \|Q - \mathbb{E}Q\|_{\text{F}}^2 \leq \frac{160d^2}{n}, \quad \|\Gamma Z\|_{\text{F}}^2 \leq 20\mathcal{A}, \quad \|\Lambda X\|_{\text{F}}^2 \leq 80\mathcal{A},$$

$\|Z\|_{\text{op}}, \|Z_0\|_{\text{op}} \leq 16$, and the conclusion of [Lemma 4.6](#). Fix such a sample.

Spectral error. By (4.3), $V^T V - I = (1 - t^2)(S - I) + t^2(\mathbb{E}Q - (I - S)) + t^2(Q - \mathbb{E}Q) + R_3 + R_4$. Since $\|X\|_{\text{op}} \leq \sqrt{2}$, $\|S\|_{\text{op}} \leq \frac{3}{2}$ and $t \leq 1$, [Lemma 4.4](#) and the event give $\|R_3\|_{\text{op}} + \|R_4\|_{\text{op}} \leq t^3(2\sqrt{40A} + 256 + 16\sqrt{20A} + \frac{3}{2} + \sqrt{160A}) \leq C_R t^3$, and therefore

$$\delta := \|V^T V - I\|_{\text{op}} \leq \eta + t^2 \frac{d^2}{n} + t^2 \sqrt{\frac{160d^2}{n}} + C_R t^3 \leq \frac{\theta}{4} t^2 + \frac{\theta}{4} t^2 + \frac{\theta}{4} t^2 + \frac{\theta}{16} t^2 < \theta t^2.$$

Graph. By (4.2), $\|V\|_{\text{op}}, \|V_0\|_{\text{op}} \leq \sqrt{2} + 16 \leq 18$ and $\|VV^T - V_0V_0^T\|_{\text{F}} \leq 36t\|Z - Z_0\|_{\text{F}}$, so $\|VV^T - V_0V_0^T\|_{\text{F}}^2 \leq C't^2d^2/n$. Let B_0 be the set of rows whose contribution to this sum exceeds κat^2 ; then $|B_0| \leq C'd/\kappa$. A row $i \notin B_0$ has at most $\kappa at^2 / ((h/2)^2 t^2 a/n) = n/50$ indices with $|(VV^T - V_0V_0^T)_{ij}| \geq \frac{h}{2} t \sqrt{a/n}$, so by [Lemma 4.6](#) it has at least $0.85n$ indices $j \neq i$ with $|(VV^T)_{ij}| \geq \frac{h}{2} t \sqrt{a/n}$.

Let $U = V(V^T V)^{-1/2}$, $P = UU^T$. By [Lemma 4.7](#) a row loses at most $6a\theta^2 t^4 / ((h/4)^2 t^2 a/n) \leq n/20$ of these indices, so each $i \notin B_0$ has at least $0.8n$ indices $j \neq i$ with $P_{ij}^2 \geq (h^2/16)at^2/n$. Also $a/2 \leq P_{ii} \leq 2a$ and $\sum_{i \in B_0} P_{ii} \leq \frac{1}{4}$. [Lemma 2.10](#) with $G = B_0^c$, part (i) with $\vartheta = 0.3$ and $\gamma = h^2 at^2/16$, and part (iii) (so $R = \frac{8}{3a}$, $F \leq \frac{4}{3}$) shows that $H(L_P) \leq \frac{7/3}{0.3\gamma} + \frac{8}{3a} \leq C_{\text{mix}}/(at^2)$.

Seed. $\|U - X\|_{\text{F}}^2 \leq 2d\delta^2 + 2t^2\|Z\|_{\text{F}}^2 \leq 42t^2d$, the diagonal error is at most $2a\delta \leq 2\theta at^2 \leq at^2/(2\Theta)$, and the barrier constant is at most $\Theta/(at^2)$. So U is a (Θ, t) -seed, and [Corollary 2.13](#) gives an ENP frame within $3\Theta t^2 d = (12\Theta/\theta)\eta d$ of X . $\square$

5 The moderate-row seed

Theorem 5.1. *There are absolute constants $A, C_m, \varepsilon_0 > 0$ such that the following holds. Let $U \in \mathbb{R}^{n \times d}$ be Parseval with $2d \leq n$, $d \geq A \log(2n)$, and $|||u_i||^2 - a| \leq \varepsilon a$ for all i , where $0 < \varepsilon \leq \varepsilon_0$. Then there is an ENP frame $\hat{U}$ with $\|\hat{U} - U\|_{\text{F}}^2 \leq C_m \varepsilon d$.*

Throughout this section $P = UU^T$, $p = \text{diag } P$, $D_p = \text{Diag}(p)$, $L = L_P$, and $\varepsilon_0 \leq \frac{1}{2}$, so $\frac{a}{2} \leq p_i \leq \frac{3a}{2} \leq \frac{3}{4}$. For a symmetric $M \in \mathbb{R}^{n \times n}$, $\mathcal{L}(M) = \sum_{i < j} M_{ij}^2 (e_i - e_j)(e_i - e_j)^T$ is the Laplacian with weights M_{ij}^2 ; thus $L = \mathcal{L}(P)$, $x^T \mathcal{L}(M) x = \frac{1}{2} \|\text{Diag } x, M\|_{\text{F}}^2$, and

$$\mathcal{L}(M) \preceq 2 \max_i \|M_i\|^2 I, \quad \mathcal{L}(M + M') \succeq \frac{1}{2} \mathcal{L}(M) - \mathcal{L}(M'), \quad (5.1)$$

by $(x_i - x_j)^2 \leq 2x_i^2 + 2x_j^2$ and $(v + w)^2 \geq \frac{1}{2}v^2 - w^2$. We write $K = nI - \mathbf{1}\mathbf{1}^T$ for the Laplacian of the complete graph.

Scales. Put $\sigma = at^2$, where $t \asymp \sqrt{\varepsilon}$ is the size of the perturbation. The perturbation creates edges of weight $\gtrsim \sigma/n$, the resulting barrier constant is $O(1/\sigma)$, and so all diagonal errors must be at most a small multiple of σ . The table lists the errors and how each is controlled ($\rho = D^2 t^2$ is the filter level, D a large constant).

diagonal error	size/ σ	reason
input	ε/t^2	$t^2 = K_0 \varepsilon$, K_0 large
linear term $2tAZ$	$6D\sqrt{\log(2n)/d}$	filter (Lemma 5.3), $d \geq A \log 2n$
quadratic fluctuation	η'	concentration, $d \geq A \log 2n$
filter bias $t^2 b_*$	0	cancelled by the drift (Lemma 5.6)
drift cross terms	$40/D + 1/D^2$	D large
base bias $t^2 b_0$	$12\varepsilon/d$	Lemma 5.6
retraction remainder	$C_E t$	t small

5.1 The horizontal tangent space and the filtered covariance

Fix an isometry $V \in \mathbb{R}^{n \times (n-d)}$ onto $(\text{im } U)^\perp$, and let $\mathcal{Z} = \mathbb{R}^{(n-d) \times d}$ with the Frobenius inner product. For $Z \in \mathcal{Z}$ the matrix $H = VZ$ satisfies $U^T H = 0$, and we put

$$Y_Z = VZU^T + UZ^T V^T, \quad \|(Y_Z)_i\|^2 = \|Z^T v_i\|^2 + \|Zu_i\|^2 \leq \|Z\|_{\text{op}}^2, \quad \|Y_Z\|_{\text{F}}^2 = 2\|Z\|_{\text{F}}^2.$$

Define

$$\mathcal{A} : \mathcal{Z} \rightarrow \mathbb{R}^n, \quad (\mathcal{A}Z)_i = v_i^T Z u_i = (HU^T)_{ii}, \quad \mathcal{A}^* x = N_x := V^T \text{Diag}(x)U.$$

Then $\mathcal{A}\mathcal{A}^* = L$, i.e. $\|N_x\|_{\text{F}}^2 = x^T L x$ (Lemma 2.2). Let

$$\bar{\mathcal{A}} = D_p^{-1/2} \mathcal{A}, \quad S = \bar{\mathcal{A}}\bar{\mathcal{A}}^* = D_p^{-1/2} L D_p^{-1/2}, \quad B = \bar{\mathcal{A}}^* \bar{\mathcal{A}}.$$

Since $L \succeq 0$ and $S = I - D_p^{-1/2}(P \circ P)D_p^{-1/2}$ with $P \circ P \succeq 0$ (Schur product theorem),

$$0 \preceq S \preceq I, \quad 0 \preceq B \preceq I, \quad \text{tr}(I - S) = \sum_i \frac{P_{ii}^2}{p_i} = d. \quad (5.2)$$

Fix an orthonormal eigenbasis of S , and for each of its members y with eigenvalue $\mu > 0$ put $f_y = D_p^{-1/2} y$ and $Z_y = \mu^{-1/2} \bar{\mathcal{A}}^* y = \mu^{-1/2} N_{f_y}$. Since $BZ_y = \mu Z_y$ and $\langle Z_y, Z_{y'} \rangle = \langle y, Sy' \rangle / \sqrt{\mu\mu'}$, the Z_y form an orthonormal eigenbasis of B on $(\ker B)^\perp$; and since $p_i \geq a/2$,

$$\|f_y\|_\infty \leq \sqrt{2/a}. \quad (5.3)$$

Definition 5.2 (Filtered covariance). For $\rho > 0$ let $C_\rho = \rho(I - B)(\rho I + B)^{-1}$ on $\mathcal{Z}$, let $Z \sim N(0, C_\rho/n)$, and $Y = Y_Z$.

Lemma 5.3. (a) $0 \preceq C_\rho \preceq I$, and $C_\rho = I$ on $\ker B$.

$$(b) \text{Cov}(\mathcal{A}Z) = n^{-1} D_p^{1/2} \rho S (I - S) (\rho I + S)^{-1} D_p^{1/2} \preceq (\rho/n) D_p.$$

$$(c) I - C_\rho = B + R_\rho \text{ with } R_\rho = \sum_{\mu>0} r_\mu Z_y \otimes Z_y, \quad r_\mu = \mu(1 - \mu)/(\rho + \mu) \geq 0, \text{ and}$$

$$\sum_{\mu>0} r_\mu \leq d, \quad \sum_{\mu>0} \frac{r_\mu}{\sqrt{\mu}} \leq \frac{d}{2\sqrt{\rho}}, \quad \sum_{\mu>0} \frac{r_\mu}{\mu} \leq \frac{d}{\rho}.$$

Proof. (a) The eigenvalues of C_ρ are $\rho(1 - \mu)/(\rho + \mu) \in [0, 1]$. (b) By singular value calculus $\bar{\mathcal{A}} C_\rho \bar{\mathcal{A}}^* = \rho S (I - S) (\rho I + S)^{-1}$, whose eigenvalues $\rho\mu(1 - \mu)/(\rho + \mu)$ are at most ρ . (c) On the μ -eigenspace, $1 - \rho(1 - \mu)/(\rho + \mu) = \mu + \mu(1 - \mu)/(\rho + \mu)$. The sums follow from $\mu/(\rho + \mu) \leq 1$, $\sqrt{\mu}/(\rho + \mu) \leq 1/(2\sqrt{\rho})$, $1/(\rho + \mu) \leq 1/\rho$, and $\sum_{\mu>0} (1 - \mu) \leq \text{tr}(I - S) = d$. $\square$

Remark 5.4. Only four properties of the filter $\phi(\mu) = \rho(1 - \mu)/(\rho + \mu)$ are used: $0 \leq \phi \leq 1 - \mu$, $\mu\phi(\mu) \leq \rho$, and the sums in (c). The hard cutoff $(1 - \mu)\mathbf{1}[\mu < \rho]$ would do as well; the rational filter keeps every object a rational function of S , which is convenient for formalisation.

5.2 Commutator identities

For $Z \in \mathcal{Z}$ define the quadratic diagonal and its polarisation

$$q_i(Z) = \|Z^T v_i\|^2 - \|Z u_i\|^2, \quad q_i(Z, M) = \langle Z^T v_i, M^T v_i \rangle - \langle Z u_i, M u_i \rangle,$$

so that $q(Z + M) = q(Z) + 2q(Z, M) + q(M)$ and, by Cauchy–Schwarz,

$$|q_i(Z, M)| \leq \|(Y_Z)_i\| \|(Y_M)_i\|, \quad |q_i(M)| \leq \|(Y_M)_i\|^2. \quad (5.4)$$

Lemma 5.5. *Let $f \in \mathbb{R}^n$, $F = \text{Diag}(f)$, and $R = I - 2P$ (an orthogonal matrix). Then:*

- (a) $q(N_f) = \mathcal{A}\Gamma_f$ with $\Gamma_f = V^T F R F U$, and $\|\Gamma_f\|_F \leq 2\|f\|_\infty \|N_f\|_F$;
- (b) $T_f := Y_{N_f} = F P + P F - 2 P F P$ satisfies $[X, T_f] = R F [X, P] + [X, P] F R$ for diagonal X ; hence $\mathcal{L}(T_f) \preceq 4\|f\|_\infty^2 L$, and $x^T \mathcal{L}(T_{e_k}) x \leq 2 \sum_j P_{kj}^2 (x_k - x_j)^2$.

Proof. (a) For $x \in \mathbb{R}^n$ put $X = \text{Diag}(x)$, $D_x = V^T X V$, $A_x = U^T X U$. Then $\langle x, q(Z) \rangle = \langle Z, D_x Z - Z A_x \rangle$ for every Z , and, since X and F commute,

$$D_x N_f - N_f A_x = V^T [X(I - P)F - F P X] U = V^T [F(I - P)X - X P F] U = D_f N_x - N_x A_f.$$

Hence $\langle x, q(N_f) \rangle = \langle N_f, D_f N_x - N_x A_f \rangle = \langle D_f N_f - N_f A_f, N_x \rangle$. As $\langle M, N_x \rangle = \sum_i x_i v_i^T M u_i = \langle \mathcal{A}M, x \rangle$ for every M , this says $q(N_f) = \mathcal{A}(D_f N_f - N_f A_f)$, and $D_f N_f - N_f A_f = V^T F(I - P)F U - V^T F P F U = \Gamma_f$. The norm bound follows from $\|D_f\|_{\text{op}}, \|A_f\|_{\text{op}} \leq \|f\|_\infty$.

(b) $Y_{N_f} = V V^T F U U^T + U U^T F V V^T = (I - P)F P + P F (I - P)$. Since X commutes with F , $[X, F P] = F[X, P]$, $[X, P F] = [X, P]F$ and $[X, P F P] = [X, P]F P + P F[X, P]$, which gives the commutator formula. Then $\|[X, T_f]\|_F \leq 2\|f\|_\infty \|[X, P]\|_F$, i.e. the first bound (Lemma 2.2). For $f = e_k$, $E_k[X, P]$ and $[X, P]E_k$ are the k th row and column of $[X, P]$, each of squared norm $\sum_j P_{kj}^2 (x_k - x_j)^2$. $\square$

5.3 The second-order mean and the drift

Lemma 5.6 (Mean). $\mathbb{E}q(Z) = a\mathbf{1} - p - b_0 - \mathcal{A}m_*$, where

$$b_0 = \frac{1}{n} L(p^{-1}), \quad \|b_0\|_\infty \leq \frac{12\varepsilon}{n}, \quad m_* = \frac{1}{n} \sum_{\mu > 0} \frac{r_\mu}{\mu} \Gamma_{f_y}, \quad \|m_*\|_F^2 \leq \frac{2a}{\rho}.$$

Proof. For $G \sim N(0, I)$ on $\mathcal{Z}$, $\mathbb{E}\|G^T v_i\|^2 = d(1 - p_i)$ and $\mathbb{E}\|G u_i\|^2 = (n - d)p_i$, so the unfiltered covariance I/n contributes $a\mathbf{1} - p$. The covariance $B = \sum_j \zeta_j \otimes \zeta_j$, $\zeta_j = \bar{\mathcal{A}}^* e_j = v_j u_j^T / \sqrt{p_j}$, contributes

$$\frac{1}{n} \sum_j q_i(\zeta_j) = \frac{1}{n} \sum_j \frac{(I - P)_{ij}^2 p_j - (1 - p_j) P_{ij}^2}{p_j} = \frac{1}{n} \left(1 - \sum_j \frac{P_{ij}^2}{p_j} \right) = \frac{1}{n} (L p^{-1})_i,$$

using $\sum_j (I - P)_{ij}^2 = 1 - p_i$ and $\sum_j P_{ij}^2 = p_i$. Here $(L p^{-1})_i = \sum_j P_{ij}^2 (p_i^{-1} - p_j^{-1})$ and $|p_i^{-1} - p_j^{-1}| \leq 2\varepsilon a / ((1 - \varepsilon)a)^2 \leq 8\varepsilon/a$, so $|(L p^{-1})_i| \leq 12\varepsilon$. Finally R_ρ/n contributes $\frac{1}{n} \sum_\mu r_\mu q(Z_y) = \frac{1}{n} \sum_\mu \frac{r_\mu}{\mu} q(N_{f_y}) = \mathcal{A}m_*$ by Lemma 5.5(a), and with $\|N_{f_y}\|_F = \sqrt{\mu}$, (5.3) and Lemma 5.3(c),

$$\|m_*\|_F \leq \frac{2}{n} \sqrt{\frac{2}{a}} \sum_{\mu > 0} \frac{r_\mu}{\sqrt{\mu}} \leq \frac{2}{n} \sqrt{\frac{2}{a}} \cdot \frac{d}{2\sqrt{\rho}} = \sqrt{\frac{2a}{\rho}}. \quad \square$$

Remark 5.7 (Why a drift). The filter removes covariance from the directions Z_y , and with it their share $t^2 b_* = t^2 \mathcal{A} m_*$ of the second-order drift towards $a\mathbf{1}$. This bias need not be small in ℓ_∞ (near a direct sum joined through a few rows it is concentrated there). But it is the *first-order* diagonal change of the deterministic horizontal direction $\frac{t}{2} m_*$, of norm $\frac{t}{2} \|m_*\|_F \leq \sqrt{a/2}/D$. Adding this direction to the noise cancels the bias exactly, at the price of cross terms of size $O(\sigma/D)$. (Equivalently, $\langle x, b_* \rangle^2 \leq \frac{2a}{\rho} x^T L x$, the bound used in [5] to remove b_* afterwards by a second scaling.)

5.4 The expected graph

For a covariance $\Sigma = \sum_k \lambda_k \psi_k \otimes \psi_k$ one has $\mathbb{E}_\Sigma \mathcal{L}(Y) = \sum_k \lambda_k \mathcal{L}(Y_{\psi_k})$.

Lemma 5.8. $\mathbb{E} \mathcal{L}(Y) \succeq \frac{a}{4n} \mathbf{K} - \left(\frac{2}{n} + \frac{8}{d} + \frac{8}{\rho} \right) L$.

Proof. Unfiltered noise. For $Z_0 \sim N(0, I/n)$ and $i \neq j$, $(Y_{Z_0})_{ij} = v_i^T Z_0 u_j + v_j^T Z_0 u_i$ has variance $n^{-1} \|v_i u_j^T + v_j u_i^T\|_F^2 = n^{-1} (p_i + p_j - 2p_i p_j - 2P_{ij}^2)$. As $p \leq \frac{3}{4}$, $p_i + p_j - 2p_i p_j \geq \frac{1}{4} (p_i + p_j) \geq \frac{a}{4}$, so $\mathbb{E} \mathcal{L}(Y_{Z_0}) \succeq \frac{a}{4n} \mathbf{K} - \frac{2}{n} L$.

Base loss. $Y_{\zeta_k} = T_{e_k} / \sqrt{p_k}$, so by Lemma 5.5(b) and $p_k \geq a/2$, $\sum_k x^T \mathcal{L}(Y_{\zeta_k}) x \leq \frac{2}{a} \cdot 2 \sum_{k,j} P_{kj}^2 (x_k - x_j)^2 = \frac{8}{a} x^T L x$; the base part B/n therefore removes at most $\frac{8}{a} L$.

Residual loss. $\mathcal{L}(Y_{Z_y}) = \mu^{-1} \mathcal{L}(T_{f_y}) \preceq \frac{8}{a\mu} L$ by Lemma 5.5(b) and (5.3), so R_ρ/n removes at most $\frac{8}{an} \sum_\mu \frac{r_\mu}{\mu} L \leq \frac{8}{\rho} L$. $\square$

5.5 A good sample

Let $\ell_i = \frac{1}{n} \sum_\mu r_\mu \| (Y_{Z_y})_i \|^2$ be the variance that R_ρ/n removes from row i . Since $\|Y_{Z_y}\|_F^2 = 2$, $\sum_i \ell_i \leq \frac{2}{n} \sum_\mu r_\mu \leq 2a$. Fix $\delta_0 = 1/6400$ and let

$$\mathfrak{B} = \{i : \ell_i > \delta_0 a\}, \quad b = |\mathfrak{B}| \leq 2/\delta_0 =: b_{\max},$$

a deterministic set depending only on U and ρ . Write $\mathcal{C}(Y) = \sum_{i < j} 2P_{ij} Y_{ij} (e_i - e_j)(e_i - e_j)^T$, so that $\mathcal{L}(P + tY) = L + t\mathcal{C}(Y) + t^2 \mathcal{L}(Y)$.

Lemma 5.9. *There is an absolute $h \in (0, 1)$ and, for each $\eta' \in (0, \frac{1}{32}]$, a constant $A_{\eta'}$ such that the following holds for all $\rho > 0$ and $0 < t \leq 1$ whenever $d \geq A_{\eta'} \log(2n)$. With positive probability:*

- (S1) $\|Z\|_{\text{op}} \leq 16$, $\max_i \|Z^T v_i\|^2 \leq 3a$ and $\max_i \|Y_i\|^2 \leq 800a$;
- (S2) $\max_i |(\mathcal{A}Z)_i| \leq 3\sqrt{\rho a \log(2n)/n}$ and $\max_i |q_i(Z) - \mathbb{E} q_i(Z)| \leq \eta' a$;
- (S3) every $i \notin \mathfrak{B}$ has at least $0.95n$ indices $j \neq i$ with $|P_{ij} + tY_{ij}| \geq h t \sqrt{a/n}$;
- (S4) for every x supported on $\mathfrak{B}$,

$$|x^T \mathcal{C}(Y) x| \leq \eta' (x^T L x + a t^2 \|x\|^2), \quad |x^T (\mathcal{L}(Y) - \mathbb{E} \mathcal{L}(Y)) x| \leq \eta' a \|x\|^2.$$

Proof. Since $C_\rho \preceq I$, every linear functional $\langle Z, \Psi \rangle$ has variance at most $\|\Psi\|_F^2/n$; we use the tools of Appendix A.

(S1): The net bound (Lemma A.1(c)) with $\sigma^2 = 1/n$ gives $\mathbb{P}(\|Z\|_{\text{op}} > 16) \leq 2 \cdot 5^n e^{-8n}$. The vector $Z^T v_i$ has covariance $\preceq n^{-1} I_d$ and trace $\leq a$; Lemma A.1(a) with $u = 2a$ (so

$\min\{u^2/\|\cdot\|_F^2, u/\|\cdot\|_{\text{op}}\} \geq 2d\}$ and a union bound give $\max\|Z^T v_i\|^2 \leq 3a$ off probability $2ne^{-cd}$. Then $\|Y_i\| \leq \|Z^T v_i\| + \|Z u_i\| \leq \sqrt{3a} + 16\sqrt{3a/2}$.

(S2): $(AZ)_i$ is Gaussian with variance $\leq \rho p_i/n \leq 3\rho a/(2n)$ (Lemma 5.3(b)); a union bound gives the first claim off probability $2n(2n)^{-3}$. The form $q_i(Z) = \langle Z, M_i Z \rangle$, $M_i Z = v_i v_i^T Z - Z u_i u_i^T$, has $\|M_i\|_{\text{op}} \leq 1$ and $\|M_i\|_F^2 \leq 2(d + np_i^2) \leq 7d$; after the covariance factor $(C_\rho/n)^{1/2}$ these become $\leq 1/n$ and $\leq \sqrt{7d}/n$, and Lemma A.1(a) with $u = \eta'a$ gives failure probability $2n \exp(-c\eta'^2 d)$.

(S3): *Variance bookkeeping.* By the proof of Lemma 5.8, for $i \neq k$ with $P_{ik}^2 \leq a/16$ the unfiltered variance of Y_{ik} is at least $a/(8n)$; at most 24 indices k have $P_{ik}^2 > a/16$. Row i of T_{e_j} is $\delta_{ij} e_j^T P + P_{ij} e_j^T - 2P_{ij} e_j^T P$, of squared norm at most $3(\delta_{ij} p_j + P_{ij}^2 + 4P_{ij}^2 p_j)$, so the base part removes from row i total variance $\frac{1}{n} \sum_j \|e_j^T T_{e_j}\|^2 / p_j \leq \frac{6}{an} \cdot 5p_i \leq \frac{45}{n}$, which exceeds $\frac{a}{32n}$ at no more than $1440n/d$ positions. For $i \notin \mathfrak{B}$ the residual part removes at most $\delta_0 a$ from row i , hence more than $\frac{a}{32n}$ at no more than $32\delta_0 n$ positions. So for $d \geq 10^6$ every $i \notin \mathfrak{B}$ has at least $0.99n$ indices $k \neq i$ with $\text{Var}(Y_{ik}) \geq \frac{a}{16n}$.

Small ball. Let $\phi: \mathbb{R} \rightarrow [0, 1]$ be C^1 , equal to 1 on $[-h, h]$, vanishing outside $[-2h, 2h]$, with $|\phi'| \leq 2/h$, and for $i \notin \mathfrak{B}$ let $F_i = \frac{1}{n} \sum_{k \neq i} \phi((P_{ik} + tY_{ik})/(t\sqrt{a/n}))$. By the density bound for a Gaussian of variance $\geq \frac{at^2}{16n}$, $\mathbb{E}F_i \leq 0.01 + \frac{4h}{\sqrt{2\pi/16}} \leq 0.01 + 7h$. As a function of the row $(Y_{ik})_k$, F_i has gradient norm at most $\frac{1}{n} \cdot \frac{2}{h} \cdot \sqrt{n/a} \cdot \sqrt{n} = 2/(h\sqrt{a})$; the row depends on Z through a linear map of norm ≤ 2 , and $Z = (C_\rho/n)^{1/2} g$ with g standard. So F_i is $4/(h\sqrt{d})$ -Lipschitz in g , and Lemma A.1(b) gives $\mathbb{P}(F_i > \mathbb{E}F_i + 0.01) \leq 2 \exp(-ch^2 d)$. Take $h = 1/500$, so that $\mathbb{E}F_i + 0.01 \leq 0.04$, and a union bound over i . On the complement, at most $0.04n$ indices $k \neq i$ have $|P_{ik} + tY_{ik}| < ht\sqrt{a/n}$, so at least $n - 1 - 0.04n \geq 0.95n$ have the reverse inequality.

(S4): Only second moments are needed, because $\mathfrak{B}$ is deterministic and bounded. Let $\Pi_{\mathfrak{B}}$ be the coordinate projection onto $\mathbb{R}^{\mathfrak{B}}$, $J_{\mathfrak{B}} = L_{\mathfrak{B}\mathfrak{B}} + at^2 I_{\mathfrak{B}}$ and $\omega_{ij} = J_{\mathfrak{B}}^{-1/2} \Pi_{\mathfrak{B}}(e_i - e_j)$, so $\|\omega_{ij}\|^2 \leq 2/(at^2)$ and $\sum_{i < j} P_{ij}^2 \omega_{ij} \omega_{ij}^T = J_{\mathfrak{B}}^{-1/2} L_{\mathfrak{B}\mathfrak{B}} J_{\mathfrak{B}}^{-1/2} \preceq I$. For x supported on $\mathfrak{B}$, $tx^T \mathcal{C}(Y)x = \langle J_{\mathfrak{B}}^{1/2} x, \Xi J_{\mathfrak{B}}^{1/2} x \rangle$ with $\Xi = \sum_{i < j} 2tP_{ij} Y_{ij} \omega_{ij} \omega_{ij}^T$, and $x^T J_{\mathfrak{B}} x = x^T L x + at^2 \|x\|^2$. The coefficients $(Y_{ij})_{i < j}$ have covariance $\preceq \frac{2}{n} I$ (the map $Z \mapsto Y_Z$ is $\sqrt{2}$ times an isometry), so

$$\mathbb{E}\|\Xi\|_F^2 \leq \frac{2t^2}{n} \sum_{i < j} 4P_{ij}^2 \|\omega_{ij}\|^4 \leq \frac{16}{an} \text{tr}(J_{\mathfrak{B}}^{-1/2} L_{\mathfrak{B}\mathfrak{B}} J_{\mathfrak{B}}^{-1/2}) \leq \frac{16b}{d},$$

and Markov's inequality gives the first half off probability $16b/(d\eta'^2)$. The principal $\mathfrak{B}$ -block of $\mathcal{L}(Y) - \mathbb{E}\mathcal{L}(Y)$ has diagonal entries $\sum_{j \neq i} (Y_{ij}^2 - \mathbb{E}Y_{ij}^2)$, a centred Gaussian quadratic form of variance at most $2 \cdot \frac{2}{n} \cdot \mathbb{E}\|Y_i\|^2 \leq \frac{12a}{n}$ (Lemma A.1(a), with $\mathbb{E}\|Y_i\|^2 \leq a(1 - p_i) + (1 - a)p_i \leq 3a$), and off-diagonal entries $-(Y_{ik}^2 - \mathbb{E}Y_{ik}^2)$ of variance at most $8/n^2$. Its expected squared Frobenius norm is at most $20b^2 a/n = 20b^2 a^2/d$, and Markov gives the second half off probability $20b^2/(d\eta'^2)$.

Finally choose $A_{\eta'}$ so large that $d \geq A_{\eta'} \log(2n)$ implies $d \geq 10^6$ and all failure probabilities above are at most $\frac{1}{10}$; their sum is less than 1. $\square$

5.6 The drifted retraction

Fix $D \geq 12$ and $t \in (0, 1]$, set $\rho = D^2 t^2$, take a sample Z as in Lemma 5.9, and let m_* be as in Lemma 5.6. Define

$$W = Z + \frac{t}{2} m_*, \quad G = I + t^2 W^T W, \quad U_t = (U + tVW)G^{-1/2}, \quad P_t = U_t U_t^T.$$

Since $U^T V = 0$, $(U + tVW)^T (U + tVW) = G$ and U_t is Parseval.

Lemma 5.10. *There is an absolute C_E such that:*

(a) $\|U_t - U\|_F^2 \leq C_E t^2 d$;

(b) every row of $E = P_t - P - tY$ satisfies $\sum_j E_{ij}^2 \leq at^2(D^{-2} + C_E t^2)$;

(c) with $\|r\|_\infty \leq C_E at^3$,

$$\text{diag } P_t - a\mathbf{1} = (1 - t^2)(p - a\mathbf{1}) + 2tAZ - t^2 b_0 + t^2(q(Z) - \mathbb{E}q(Z)) + t^3 q(Z, m_*) + \frac{t^4}{4} q(m_*) + r;$$

(d) $\|\text{diag } P_t - a\mathbf{1}\|_\infty \leq at^2(\frac{\varepsilon}{t^2} + 6D\sqrt{\log(2n)/d} + \frac{12\varepsilon}{d} + \eta' + \frac{40}{D} + \frac{1}{2D^2} + C_E t)$.

Proof. By Lemma 5.6, $\frac{t}{2}\|m_*\|_{\text{op}} \leq \frac{t}{2}\|m_*\|_F \leq \sqrt{a/2}/D \leq \frac{1}{24}$, so with (S1): $\|W\|_{\text{op}} \leq 17$, $\|W\|_F \leq 17\sqrt{d}$, $\|W^T v_i\| \leq 2\sqrt{a}$, $\|W u_i\| \leq 17\sqrt{3a/2}$, and $\|(Y_{m_*})_i\|^2 \leq \|m_*\|_{\text{op}}^2 \leq 2a/\rho$. Hence $\|G^{-1} - I\|_{\text{op}}$ and $\|G^{-1/2} - I\|_{\text{op}}$ are at most $289t^2$, $\|U + tVW\|_{\text{op}} \leq 18$ and $\|u_i + tW^T v_i\| \leq 4\sqrt{a}$.

(a) $\|U_t - U\|_F \leq \|U + tVW\|_F \|G^{-1/2} - I\|_{\text{op}} + t\|W\|_F \leq Ct\sqrt{d}$.

(b) $E = \frac{t^2}{2} Y_{m_*} + E'$ with $E' = P_t - P - tY_W = (U + tVW)(G^{-1} - I)(U + tVW)^T + t^2 V W W^T V^T$, whose rows have norm at most $Ct^2\sqrt{a}$. The rows of $\frac{t^2}{2} Y_{m_*}$ have squared norm at most $\frac{t^4}{4} \cdot \frac{2a}{\rho} = \frac{at^2}{2D^2}$. Use $(\alpha + \beta)^2 \leq 2\alpha^2 + 2\beta^2$.

(c) Let $x_i = u_i + tW^T v_i$, the i th row of $U + tVW$. Inserting $G^{-1} = I - t^2 W^T W + t^4 (W^T W)^2 G^{-1}$ into $(P_t)_{ii} = x_i^T G^{-1} x_i$ and using $u_i^T W^T v_i = (AW)_i$,

$$(P_t)_{ii} = p_i + 2t(AW)_i + t^2 q_i(W) - 2t^3 \langle W u_i, W W^T v_i \rangle - t^4 \|W W^T v_i\|^2 + t^4 x_i^T (W^T W)^2 G^{-1} x_i,$$

and the last three terms are $O(at^3)$. Now $2tAW = 2tAZ + t^2 Am_*$, $q(W) = q(Z) + tq(Z, m_*) + \frac{t^2}{4} q(m_*)$, and $\mathbb{E}q(Z) = a\mathbf{1} - p - b_0 - Am_*$ by Lemma 5.6; the terms $\pm t^2 Am_*$ cancel.

(d) In (c): $(1 - t^2)\|p - a\mathbf{1}\|_\infty \leq \varepsilon a$; $2t\|AZ\|_\infty \leq 6tDt\sqrt{a \log(2n)/n}$ by (S2), which is $6Dat^2\sqrt{\log(2n)/d}$; $t^2\|b_0\|_\infty \leq 12\varepsilon t^2/n$; the fluctuation is at most $\eta' at^2$; by (5.4) and (S1), $t^3|q_i(Z, m_*)| \leq t^3\sqrt{800a}\sqrt{2a/\rho} = 40at^2/D$ and $\frac{t^4}{4}|q_i(m_*)| \leq \frac{t^4}{4} \cdot \frac{2a}{\rho} = \frac{at^2}{2D^2}$. $\square$

Lemma 5.11 (Graph of the seed). *There are absolute $t_* \in (0, 1]$ and C_{core} such that if $\eta' \leq \frac{1}{32}$, $D \geq \max\{12, 13/h\}$, $b \leq n/10$ and $t \leq t_*$, then $H(L_{P_t}) \leq C_{\text{core}}/(at^2)$.*

Proof. Core. In a row $i \notin \mathfrak{B}$, by Lemma 5.10(b), at most $at^2(D^{-2} + C_E t^2)/(\frac{h^2}{4} \frac{at^2}{n}) \leq 0.05n$ indices have $|E_{ij}| \geq \frac{h}{2} t\sqrt{a/n}$, once $D \geq 13/h$ and $t^2 \leq h^2/(160C_E)$. With (S3), every $i \notin \mathfrak{B}$ has at least $0.9n$ indices $j \neq i$ with $(P_t)_{ij}^2 \geq \frac{h^2}{4} \frac{at^2}{n}$.

Block. For x supported on $\mathfrak{B}$, Lemma 5.8, $x^T K x \geq (n - b)\|x\|^2$ and (S4) give

$$x^T \mathcal{L}(P + tY)x \geq \left(1 - \eta' - t^2\left(\frac{2}{n} + \frac{8}{d}\right) - \frac{8}{D^2}\right)x^T Lx + at^2\left(\frac{1}{4}\left(1 - \frac{b}{n}\right) - 2\eta'\right)\|x\|^2 \geq \frac{at^2}{8}\|x\|^2.$$

By (5.1) and Lemma 5.10(b), $L_{P_t} = \mathcal{L}(P + tY + E) \succeq \frac{1}{2}\mathcal{L}(P + tY) - 2at^2(D^{-2} + C_E t^2)I$, which is at least $\frac{at^2}{32}$ on vectors supported on $\mathfrak{B}$ once $t^2 \leq 1/(128C_E)$.

Barrier. Lemma 2.10 with $G = \mathfrak{B}^c$, part (i) with $\vartheta = 0.4$, $\gamma = h^2 at^2/4$, and part (ii) with $\lambda = at^2/32$, gives $H(L_{P_t}) \leq \frac{1+b}{0.1h^2 at^2} + \frac{32\sqrt{b}}{at^2}$. Put $C_{\text{core}} = 10(1 + b_{\text{max}})/h^2 + 32\sqrt{b_{\text{max}}}$. $\square$

5.7 Proof of Theorem 5.1

Constants. The numbers $\delta_0, b_{\max}, h, C_E, t_*, C_{\text{core}}$ are absolute and do not depend on D (for $D \geq 12$). Put $\Theta = \max\{2, C_{\text{core}}, C_E\}$ and choose, in this order: $D \geq \max\{13/h, 500\Theta\}$, so that $\frac{40}{D} + \frac{1}{2D^2} \leq \frac{1}{12\Theta}$; $K_0 = 12\Theta$ and $\eta' = \min\{\frac{1}{32}, \frac{1}{12\Theta}\}$; $A \geq A_{\eta'}$ so large that $d \geq A \log(2n)$ implies $6D\sqrt{\log(2n)}/d \leq \frac{1}{12\Theta}$ and $b_{\max} \leq n/10$; and $\varepsilon_0 \leq \frac{1}{2}$ so that $t = \sqrt{K_0\varepsilon}$ satisfies $t \leq t_*$, $C_E t \leq \frac{1}{12\Theta}$ and $12\varepsilon \leq \frac{1}{12\Theta}$ for $\varepsilon \leq \varepsilon_0$. None of these choices depends on n, d, ε or U .

Seed. With $t^2 = K_0\varepsilon$, Lemma 5.10(d) bounds the diagonal error of P_t by $at^2 \cdot \frac{6}{12\Theta} = \frac{at^2}{2\Theta}$; Lemma 5.11 bounds the barrier constant by $\Theta/(at^2)$; and Lemma 5.10(a) bounds $\|U_t - U\|_{\text{F}}^2$ by $\Theta t^2 d$. So U_t is a (Θ, t) -seed for U , and Corollary 2.13 gives an ENP frame $\hat{U}$ with $\|\hat{U} - U\|_{\text{F}}^2 \leq 3\Theta t^2 d = 36\Theta^2 \varepsilon d$. $\square$

Remark 5.12. Compared with [5], the drift replaces the second minimisation that removed $t^2 b_*$ (and with it the Hessian comparison under scaling, the global two-sided comparison $L_{P_t} \asymp L + at^2 I$, the concentration of the full squared Gaussian graph and the matrix-series bound for the cross term, and the re-verification of the dense core after the correction). The only spectral input left is the gap on the bounded deterministic block $\mathfrak{B}$, which needs second moments only.

6 Assembly

We use the two seeds through the following equal-norm formulation.

Proposition 6.1. *There is an absolute constant C_0 such that every $X \in \mathbb{R}^{n \times d}$ with $1 \leq d$, $2d \leq n$ and $\|x_i\|^2 = a$ for all i admits an ENP frame W with*

$$\|X - W\|_{\text{F}}^2 \leq C_0 d \|X^T X - I\|_{\text{op}}.$$

Proof. Let $\eta = \|X^T X - I\|_{\text{op}}$; if $\eta = 0$ take $W = X$. Let A, ε_0, C_m be the constants of Theorem 5.1 and B, c_*, C_* those of Theorem 4.1, and fix an integer $D \geq 2$ with $A \log(2Bd^2) \leq d$ for all $d \geq D$.

Case $d < D$. Lemma 3.1 gives $\|X - W\|_{\text{F}}^2 \leq \eta d(d-1) \leq D\eta d$.

Case $d \geq D$, $n \geq Bd^2$. If $\eta \leq c_*$ use Theorem 4.1; otherwise Corollary 3.2 gives cost $4d \leq (4/c_*)\eta d$.

Case $d \geq D$, $n < Bd^2$. Then $A \log(2n) \leq d$. If $\eta \leq \varepsilon_0/4$, let $U = X(X^T X)^{-1/2}$. The singular values of X lie in $[\sqrt{1-\eta}, \sqrt{1+\eta}]$, so $\|X - U\|_{\text{F}}^2 \leq d\eta^2$, and $\|u_i\|^2 = x_i^T (X^T X)^{-1} x_i \in [a/(1+\eta), a/(1-\eta)]$, so $|\|u_i\|^2 - a| \leq 2\eta a$. Theorem 5.1 with $\varepsilon = 2\eta$ gives an ENP W with $\|U - W\|_{\text{F}}^2 \leq 2C_m \eta d$, and $\|X - W\|_{\text{F}}^2 \leq 2d\eta^2 + 4C_m \eta d$. If $\eta > \varepsilon_0/4$ use the trivial cost $4d$. $\square$

Proof of Theorem 1.2. Replacing (P, Q) by $(I - P, I - Q)$ preserves β , ranks of complements and distances, so we may assume $d \leq n/2$; the case $d = 0$ is trivial. Write $P = UU^T$ with $U^T U = I$, $p_i = \|u_i\|^2$ and $\delta = \beta/a$. If $\delta \geq \frac{1}{2}$, any ENP W (Corollary 3.2) gives $\|P - WW^T\|_{\text{F}}^2 \leq 2d \leq 4n\beta$. Otherwise let $x_i = \sqrt{a} u_i / \|u_i\|$. Then $\|X - U\|_{\text{F}}^2 = \sum_i (\sqrt{p_i} - \sqrt{a})^2 \leq \sum_i (p_i - a)^2 / a \leq \delta^2 d$ and $X^T X = U^T \text{Diag}(a/p) U$, so $\|X^T X - I\|_{\text{op}} \leq \max_i |a/p_i - 1| \leq \delta/(1-\delta) \leq 2\delta$. Proposition 6.1 gives an ENP W with $\|U - W\|_{\text{F}}^2 \leq 2\delta^2 d + 4C_0 \delta d \leq (1 + 4C_0)n\beta$, using $\delta d = n\beta$. Take $Q = WW^T$ and use Lemma 2.1. $\square$

Proof of Theorem 1.1. If $\varepsilon \geq \frac{1}{2}$, an ENP W gives $\|X - W\|_{\text{F}}^2 \leq 2\|X\|_{\text{F}}^2 + 2d \leq 6d \leq 12\varepsilon d$, since $\|X\|_{\text{F}}^2 = \text{tr } X^T X \leq (1 + \varepsilon)d$. Let $\varepsilon < \frac{1}{2}$ and $U = X(X^T X)^{-1/2}$, $P = UU^T$. As above $\|X - U\|_{\text{F}}^2 \leq \varepsilon^2 d$ and $P_{ii} = x_i^T (X^T X)^{-1} x_i \in [\frac{1-\varepsilon}{1+\varepsilon}a, \frac{1+\varepsilon}{1-\varepsilon}a]$, so $\beta = \max_i |P_{ii} - a| \leq 4\varepsilon a$. Theorem 1.2 gives Q with $\|P - Q\|_{\text{F}}^2 \leq 4C\varepsilon d$, and by Lemma 2.1 there is an isometry W spanning Q —hence ENP—with $\|U - W\|_{\text{F}}^2 \leq \|P - Q\|_{\text{F}}^2$. Then $\|X - W\|_{\text{F}}^2 \leq 2\varepsilon^2 d + 8C\varepsilon d$. $\square$

A Gaussian tools

Lemma A.1. *The following hold with absolute constants.*

(a) (Quadratic forms.) *For g standard Gaussian and A symmetric, $\text{Var}(g^T A g) = 2\|A\|_F^2$ and*

$$\mathbb{P}\{|g^T A g - \text{tr } A| > u\} \leq 2 \exp\left[-\frac{1}{8} \min\left\{\frac{u^2}{\|A\|_F^2}, \frac{u}{\|A\|_{\text{op}}}\right\}\right].$$

For a Gaussian vector $\zeta = \Sigma^{1/2}g$, apply this to $\Sigma^{1/2}A\Sigma^{1/2}$, whose norms are at most $\|\Sigma\|_{\text{op}}$ times those of A .

(b) (Lipschitz functions.) *If F is C^1 with $\|\nabla F\| \leq \ell$ and g is standard Gaussian, then $\mathbb{P}\{|F(g) - \mathbb{E}F(g)| > u\} \leq 2 \exp(-2u^2/(\pi^2 \ell^2))$.*

(c) (Nets.) *If Z is an $r \times s$ centred Gaussian matrix with $\text{Cov}(\text{vec } Z) \preceq \sigma^2 I$, then $\mathbb{P}\{\|Z\|_{\text{op}} > u\} \leq 2 \cdot 5^{r+s} \exp(-u^2/(32\sigma^2))$.*

Proof. (a) Diagonalise A with eigenvalues λ_j . For $|s| \leq 1/(4\|A\|_{\text{op}})$, $\log \mathbb{E}e^{s(g^T A g - \text{tr } A)} = \sum_j [-s\lambda_j - \frac{1}{2} \log(1 - 2s\lambda_j)] \leq 2s^2\|A\|_F^2$, since $-x - \frac{1}{2} \log(1 - 2x) = \sum_{k \geq 2} (2x)^k/(2k) \leq 2x^2$ for $|x| \leq \frac{1}{4}$. The exponential Markov inequality with $s = \min\{u/(4\|A\|_F^2), 1/(4\|A\|_{\text{op}})\}$, applied to both signs, gives the tail. The variance follows from $\mathbb{E}g_j^4 = 3$.

(b) Let g' be an independent copy, $g_\theta = g \cos \theta + g' \sin \theta$ and $\dot{g}_\theta = -g \sin \theta + g' \cos \theta$; for each θ the pair $(g_\theta, \dot{g}_\theta)$ is standard Gaussian. By the fundamental theorem of calculus and Jensen on $[0, \pi/2]$,

$$\mathbb{E}e^{s(F(g') - F(g))} \leq \frac{2}{\pi} \int_0^{\pi/2} \mathbb{E} \exp\left[\frac{\pi s}{2} \langle \nabla F(g_\theta), \dot{g}_\theta \rangle\right] d\theta \leq e^{\pi^2 s^2 \ell^2 / 8}.$$

Jensen in g' gives $\mathbb{E}e^{s(F(g) - \mathbb{E}F)} \leq e^{\pi^2 s^2 \ell^2 / 8}$; optimise in s .

(c) Let $\mathcal{N}_1, \mathcal{N}_2$ be $\frac{1}{2}$ -nets of the unit spheres of $\mathbb{R}^r$ and $\mathbb{R}^s$ with at most $5^r, 5^s$ points. For a vector w , a net point y' within $\frac{1}{2}$ of $w/\|w\|$ has $\langle w, y' \rangle \geq \frac{1}{2}\|w\|$; applying this twice gives $\|Z\|_{\text{op}} \leq 4 \max_{x \in \mathcal{N}_1, y \in \mathcal{N}_2} |x^T Z y|$. Each $x^T Z y$ is Gaussian with variance $\leq \sigma^2$; a union bound finishes. $\square$

Covariance domination. If a Gaussian vector ξ has covariance $\Gamma \preceq \gamma I$ and (M_α) are matrices, then $\mathbb{E}\|\sum_\alpha \xi_\alpha M_\alpha\|_F^2 = \text{tr}(\Gamma G) \leq \gamma \sum_\alpha \|M_\alpha\|_F^2$, where $G \succeq 0$ is the Gram matrix of the M_α . This is the only form in which correlated coefficients enter (Lemmas 4.3 and 5.9).

B Relation to the original manuscript and its formalisation

The overall strategy—perturb by a Gaussian tangent of size $\sqrt{\varepsilon}$, then balance by a static scaling controlled in ℓ_∞ ; two seeds for $d \gtrsim \log n$ and $n \gtrsim d^2$; Hamilton–Moitra for bounded d —and most individual estimates are those of [5]. The differences are as follows.

1. *Bias removal by a drift.* In [5] the filter bias $t^2 b_*$ is controlled through $\langle x, b_* \rangle^2 \leq (2a/\rho)x^T L x$ and removed after the perturbation by a second, local scaling (a minimisation on a mean-zero ℓ_∞ -ball, with a Hessian comparison under scaling). This requires the two-sided comparison $L_{P_t} \asymp L + at^2 I$ on all of $\mathbf{1}^\perp$, hence the concentration of the full squared Gaussian graph, a matrix-series bound for the cross term, and a re-verification of the dense core after the

correction. Here the identity $q(N_f) = \mathcal{A}(V^T F(I - 2P)FU)$ (Lemma 5.5) exhibits $b_* = \mathcal{A}m_*$ with $\|m_*\|_F^2 \leq 2a/\rho$, and the drift $\frac{t}{2}m_*$ added to the noise cancels the bias at first order (Lemmas 5.6 and 5.10). The only spectral input left is a gap on a bounded deterministic set of rows, proved with second moments (Lemma 5.9(S4)). This observation was found by one of the review agents used in preparing this text.

2. *Many-row seed for all small errors.* In [5] the many-row theorem assumes $\eta \leq c/d^2$, because the cubic remainder of the normalisation identity is bounded termwise by $O(t^3 d)$, and the operator norm of the noise by $t^2 \|Z\|_F^2 \leq 1$. Lemma 4.4 uses the first-order constraint $X^T Z + Z^T X = 0$ a second time to centre the weights, giving $O(t^3)$, and the net bound controls $\|Z\|_{\text{op}}$. Theorem 4.1 therefore holds for $\eta \leq c_*$.
3. *No block partition.* With 2., the global argument needs no random partition into blocks of size $\lceil Ld^6 \rceil$ (and no uniform-subset variance identity): the regimes $d < D$, $n \geq Bd^2$, $n < Bd^2$ of Proposition 6.1 cover everything, since $A \log(2Bd^2) \leq d$ for $d \geq D$.
4. *Barrier constant.* The ℓ_∞ Poisson constant is replaced by the half-set hitting time (Definition 2.6), which is what the median barrier actually uses (the two agree up to a factor 2, Remark 2.11). With it the maximum principle needs no connectivity argument, and the dense-core estimate becomes one explicit barrier (Lemma 2.10) instead of a Schur-complement argument with an overlap condition. The balancing constant improves from 8 to $\frac{15}{4}$.
5. *Presentation.* One toolbox section; both seeds routed through Definition 2.12 and Corollary 2.13; a table of scales for the moderate seed; complete proofs of all sampling estimates, including the variance bookkeeping behind the dense core; explicit, acyclic constant choices.

Correctness of the original. An independent section-by-section check of [5] (by hand and numerically) found no false statement and no substantive gap; the issues found were notational overloading, terse justifications, and harmless constant slips (e.g. $\mathcal{L}(W^2) \preceq 4p_{\max}L$ where $2p_{\max}L$ is stated).

Formal status. The Lean 4 development accompanying [5] proves Theorems 1.1 and 1.2 by the original route. The arguments of Sections 4 to 6 and the barrier formulation of Section 2 given here are *not* formalised; Section 3 and the remaining parts of Section 2 follow the formalised proof up to constants and presentation.

References

- [1] J. Cahill and P. G. Casazza. *The Paulsen problem in operator theory*. Operators and Matrices 7 (2013). [arXiv:1102.2344](#).
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- [4] L. C. Lau and A. Ramachandran. *Optimal bounds for Tyler’s M-estimator for elliptical distributions*, Section 7. [arXiv:2510.13751](#), 2025.

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